



Exelon Generation.

10CFR 50.73

CCN: 18-133

December 21, 2018

U. S. Nuclear Regulatory Commission
ATTN: Document Control Desk
Washington, DC 20555-0001

Peach Bottom Atomic Power Station (PBAPS) Unit 2
Renewed Facility Operating License No. DPR-44
NRC Docket No. 50-277

Subject: Licensee Event Report (LER) 2-17-001, Revision 1

Enclosed is a revision to Licensee Event Report 2-17-001 concerning non-conforming conditions associated with tornado missile protection. In accordance with NEI 99-04, the regulatory commitment contained in this correspondence is to restore compliance with the regulations. The specific methods that have been planned to restore and maintain compliance are discussed in the LER. If you have any questions or require additional information, please do not hesitate to contact Matt Retzer at 717-456-4351.

Sincerely,

Matthew J. Herr
Plant Manager
Peach Bottom Atomic Power Station

MJH/dnd/IR 4196558

Enclosure

cc: US NRC, Administrator, Region I
US NRC, Senior Resident Inspector
R. R. Janati, Commonwealth of Pennsylvania
D. Tancabel, State of Maryland
B. Watkins, PSE&G, Financial Controls and Co-owner Affairs

IE22
NRR

**LICENSEE EVENT REPORT (LER)**

(See Page 2 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Privacy and Information Collections Branch (T-5 F53), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollects.Resource@nrc.gov, and to the Desk Officer, Office of Information and Regulatory Affairs, NEOB-10202, (3150-0104), Office of Management and Budget, Washington, DC 20503. If a means used to impose an information collection does not display a currently valid OMB control number, the NRC may not conduct or sponsor, and a person is not required to respond to, the information collection.

1. FACILITY NAME

Peach Bottom Atomic Power Station Unit 2

2. DOCKET NUMBER

05000277

3. PAGE

1 OF 4

4. TITLE

Emergency Diesel Generator Exhaust Stacks Nonconforming Design for Tornado Missile Protection

5. EVENT DATE			6. LER NUMBER			7. REPORT DATE			8. OTHER FACILITIES INVOLVED	
MONTH	DAY	YEAR	YEAR	SEQUENTIAL NUMBER	REV NO.	MONTH	DAY	YEAR	FACILITY NAME	DOCKET NUMBER
01	09	2017	2017	- 001	- 01	12	21	18	Peach Bottom APS Unit 3	05000278
									FACILITY NAME	DOCKET NUMBER

9. OPERATING MODE	11. THIS REPORT IS SUBMITTED PURSUANT TO THE REQUIREMENTS OF 10 CFR §: (Check all that apply)			
1	☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)
	☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☒ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)
	☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)
	☐ 20.2203(a)(2)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)
10. POWER LEVEL 100%	☐ 20.2203(a)(2)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(v)(A)	☐ 73.71(a)(4)
	☐ 20.2203(a)(2)(iii)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(v)(B)	☐ 73.71(a)(5)
	☐ 20.2203(a)(2)(iv)	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(C)	☐ 73.77(a)(1)
	☐ 20.2203(a)(2)(v)	☐ 50.73(a)(2)(i)(A)	☒ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(i)
	☐ 20.2203(a)(2)(vi)	☒ 50.73(a)(2)(i)(B)	☒ 50.73(a)(2)(vii)	☐ 73.77(a)(2)(ii)
	☐ 50.73(a)(2)(i)(C)	☐ OTHER	Specify in Abstract below or in NRC Form 366A	

12. LICENSEE CONTACT FOR THIS LER

LICENSEE CONTACT

Matthew Retzer, Regulatory Assurance Manager

TELEPHONE NUMBER (Include Area Code)

717-456-4351

13. COMPLETE ONE LINE FOR EACH COMPONENT FAILURE DESCRIBED IN THIS REPORT

CAUSE	SYSTEM	COMPONENT	MANU-FACTURER	REPORTABLE TO EPIX	CAUSE	SYSTEM	COMPONENT	MANU-FACTURER	REPORTABLE TO EPIX

14. SUPPLEMENTAL REPORT EXPECTED☐ YES (If yes, complete 15. EXPECTED SUBMISSION DATE) ☒ NO**15. EXPECTED SUBMISSION DATE**

MONTH DAY YEAR

ABSTRACT (Limit to 1400 spaces, i.e., approximately 15 single-spaced typewritten lines)

On 1/9/17, it was determined that the site's Emergency Diesel Generators do not conform with the licensing basis for protection against tornado generated missiles. The exhaust stacks for the four on-site diesel generators extend approximately seven feet above the roof of the diesel generator building. In the event of a tornado, debris generated from the tornado could strike the exhaust stacks and, if at a sufficient mass and velocity, could crimp the exhaust stacks in a manner that would affect diesel generator operation. An extent of condition review identified unprotected cables that could be impacted by a tornado and affect the ability to maintain safe shutdown.

As a result of the non-conforming conditions, on 1/9/17 at 1530, all four emergency diesel generators were declared inoperable and on 11/1/18, the Reactor Core Isolation Cooling system and the RHR Suppression Pool Cooling system were declared inoperable. Compensatory measures were put in place and, in accordance with NRC guidance contained in EGM 15-002, the equipment was returned to an operable but non-conforming status.

Long-term corrective actions are being evaluated in accordance with EGM 15-002 and the plant's licensing basis. There are no actual consequences as a result of the non-conforming condition.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Privacy and Information Collections Branch (T-5 F53), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollections.Resource@nrc.gov, and to the Desk Officer, Office of Information and Regulatory Affairs, NEOB-10202, (3150-0104), Office of Management and Budget, Washington, DC 20503. If a means used to impose an information collection does not display a currently valid OMB control number, the NRC may not conduct or sponsor, and a person is not required to respond to, the information collection.

1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER		
		YEAR	SEQUENTIAL NUMBER	REV NO.
Peach Bottom Atomic Power Station Unit 2	05000-277	2017	- 001	- 01

NARRATIVE**Unit Conditions Prior to the Event**

Unit 2 and Unit 3 were both operating in Mode 1 at approximately 100% rated thermal power. There were no structures, systems or components out of service that contributed to this event.

Description of Event

On 1/9/17, based on an engineering analysis, it was determined that the site's four Emergency Diesel Generators [EIS: DG] do not conform with the licensing basis for protection against tornado generated missiles. As a result, one or more of the diesel generators may not have been able to perform their safety design function if a tornado were to impact the site and if debris of sufficient mass and velocity were to strike the diesel engine exhaust stacks.

As a result of the non-conforming condition, on 1/9/17, at 1530, all four emergency diesel generators were declared inoperable in accordance with Technical Specification (TS) 3.8.1. With two or more diesel generators inoperable, TS 3.8.1 Condition F requires all but one to be restored to an operable status within two hours, or per Condition G, be in Mode 3 in 12 hours. Compensatory measures were verified as being in place in accordance with NRC guidance contained in EGM 15-002, "Enforcement Discretion for Tornado-Generated Missile Protection Noncompliance" (ML15111A269). As allowed by the EGM, the Emergency Diesel Generators were immediately returned to an operable but non-conforming status.

Following identification of the EDG stack vulnerability, a walkdown of the site was performed to identify any other non-conforming conditions. Three rooms that contain safety-related cables associated with equipment needed for safe shutdown following a tornado were identified as non-conforming. Two of the rooms ("MG-Set rooms"), adjacent to each unit's reactor building, have an exterior wall that is steel panel on I-beam construction, which does not conform with the site's licensing basis. The third room, a fan room in the Radwaste Building, between the two MG-Set rooms, has an exterior wall that consists of louvers that do not provide adequate protection from tornado generated missiles.

As a result of the non-conforming conditions that affect equipment required for safe shutdown following a tornado, on 11/1/2018, the Reactor Core Isolation Cooling (RCIC) [EIS:BN] system and the RHR Suppression Pool Cooling [EIS:BT] system were declared inoperable for both units. Technical Specification (TS) 3.5.3, Condition A was entered for an inoperable RCIC system and TS 3.6.2.3, Condition C was entered for two inoperable RHR Suppression Pool Cooling subsystems. Compensatory measures described in EGM 15-002 were verified as being in place and, as allowed by the EGM, the systems were immediately returned to an operable but non-conforming status.

EGM 15-002, along with Interim Staff Guidance DSS-ISG-2016-01, "Clarification of Licensee Actions in Receipt of Enforcement Discretion Per Enforcement Guidance Memorandum EGM 15-002" (ML15348A202), provide guidance from the NRC for acceptable compensatory measures that licensees can put in place if equipment is determined to be inoperable as a result of tornado generated missiles.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Privacy and Information Collections Branch (T-5 F53), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollects.Resource@nrc.gov, and to the Desk Officer, Office of Information and Regulatory Affairs, NEOB-10202, (3150-0104), Office of Management and Budget, Washington, DC 20503. If a means used to impose an information collection does not display a currently valid OMB control number, the NRC may not conduct or sponsor, and a person is not required to respond to, the information collection.

1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER		
		YEAR	SEQUENTIAL NUMBER	REV NO.
Peach Bottom Atomic Power Station Unit 2	05000-277	2017	- 001	- 01

NARRATIVE

Compensatory measures were verified as being in place for both conditions in accordance with EGM 15-002.

Analysis of the Event

Four emergency diesel generators are located on-site which provide backup power in the event of a loss of all offsite power. Each diesel generator is rated at 2,600 kW for continuous operation at 4,160 volts. The diesel generators provide sufficient power to safely shut down both units in the event of a design basis accident in one unit and a failure of one of the diesel generators. Each diesel generator is housed in a seismic Class I structure and designed to be protected against earthquakes, floods, tornadoes and other natural phenomena. However, the diesel engine exhaust stacks, which extend approximately seven feet above the roof of the building, are not protected against tornado generated missiles. The exhaust stacks are 24-inch diameter steel pipes with a 3/8-inch wall thickness. It was determined that, if the stacks were struck by a design basis tornado missile, they could be crimped in such a manner as to prevent the diesel generator from performing its design function. The design basis tornado missile for the site includes a wood plank that is 4 inches thick, 12 inches wide and 12 feet long traveling at 300 mph.

The site's Updated Final Safety Analysis Report (UFSAR) states that a tornado traversing the site will not impair the ability to shut down the reactor, the integrity of primary containment and the capability of essential heat removal systems. It states that components which directly affect the ultimate safe shutdown of the plant are located either in reinforced concrete structures or underground for tornado protection. In the event of a tornado on or near the site, damage could occur to transmission lines or associated equipment, resulting in an automatic reactor shutdown and a loss of offsite power. If this were to occur, there are several methods that are available for maintaining the plant in a safe shutdown condition. As a minimum, RCIC would be used to maintain water level in the reactor vessel and RHR Suppression Pool Cooling would be used to remove decay heat. Cables for both of these systems were identified as being in the MG-Set rooms or the fan room and therefore vulnerable to tornado missiles.

There were no actual consequences resulting from this non-conforming condition. A tornado that could produce the design basis missile is highly unlikely in the mid-Atlantic region. The probability of missiles causing more than one diesel generator to become inoperable or damaging specific cables resulting in RCIC and RHR Suppression Pool Cooling to become inoperable is also very low. EGM 15-002 describes an NRC generic bounding risk analysis for tornado missile protection non-compliances, which determined that broad enforcement discretion of up to five years will not impose any significant additional risk to the public health and safety.

This condition is being reported in accordance with the following criteria:

- 10CFR 50.73(a)(2)(i)(B) for any condition which was prohibited by the plant's Technical Specifications.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Privacy and Information Collections Branch (T-5 F53), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollects.Resource@nrc.gov, and to the Desk Officer, Office of Information and Regulatory Affairs, NEOF-10202, (3150-0104), Office of Management and Budget, Washington, DC 20503. If a means used to impose an information collection does not display a currently valid OMB control number, the NRC may not conduct or sponsor, and a person is not required to respond to, the information collection.

1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER		
		YEAR	SEQUENTIAL NUMBER	REV NO.
Peach Bottom Atomic Power Station Unit 2	05000-277	2017	- 001	- 01

NARRATIVE

- 10CFR 50.73(a)(2)(ii)(B) for any event or condition that resulted in the nuclear power plant being in an unanalyzed condition that significantly degraded plant safety.
- 10CFR 50.73(a)(2)(v)(D) for any condition that could have prevented the fulfillment of the safety function of systems that are needed to mitigate the consequences of an accident.
- 10CFR 50.73(a)(2)(vii)(D) for any event where a single cause or condition caused at least two independent trains to become inoperable in a single system designed to mitigate the consequences of an accident.

Cause of the Event

The design of the EDG exhaust stacks was part of the original plant design and has existed since the initial construction and licensing of the plant. Peach Bottom Unit 2 and Unit 3 received their operating licenses in 1973 and 1974, respectively, based in part on the design of the emergency diesel generators as described in the Final Safety Analysis Report (FSAR). This report did not address the fact that the exhaust stacks are not protected against tornado missiles and no other documented engineering analysis has been identified. It is not known if this was not evaluated at the time of the original design and licensing process, or if it was considered acceptable due to the low likelihood of damage to the exhaust stacks from tornado missiles.

Past perception of this condition was that only one tornado generated missile needed to be considered and only one diesel exhaust stack would be impacted. Since only three emergency diesel generators are needed to safely shut down the plant, such a condition was considered to be included in the licensing basis of the plant. Additional industry and NRC information on this issue pointed to the need to have all diesel generators survive a tornado with the capability to perform their safety function intact.

It is not known if the original plant design included routing of cables for RCIC and RHR Suppression Pool Cooling through the MG-Set and/or fan rooms. Since the plant was originally licensed, plant modifications have been performed that have affected cable routing in these rooms. For example, alternative shutdown panels were installed in the MG-Set rooms in the mid-1980's in response to new fire protection regulations (10CFR50 Appendix R).

Corrective Actions

Compensatory measures were implemented in accordance with NRC guidance contained in EGM 15-002 and DSS-ISG-2016-01 as described earlier. Enforcement discretion provided by EGM 15-002 remains in effect for five years for Peach Bottom (until June 2020), at which time modifications, license amendments or other actions must be implemented to fully resolve the issue. Additional evaluation of these options is currently being conducted.

Previous Similar Occurrences

No previous similar events have occurred at the site.